FL-EHDS: A Privacy-Preserving Federated Learning Framework for the European Health Data Space

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***Abstract*—The European Health Data Space (EHDS), established by Regulation (EU) 2025/327, mandates cross-border health data analytics while preserving citizen privacy. We present FL-EHDS, a three-layer compliance framework integrating EHDS governance (Health Data Access Bodies, data permits, Article 71 opt-out) with federated learning orchestration (17 algorithms including ICML/ICLR 2024–2025 advances, differential privacy, secure aggregation) and data holder components (FHIR R4 preprocessing). Validation across 4,241+ experiments on tabular clinical and medical imaging datasets demonstrates that *personalization architecture—not aggregation algorithm—is the critical deployment decision*: only methods maintaining separate local models (Ditto, HPFL) differentiate from FedAvg on compact clinical models, producing up to 12.6pp accuracy gains. Privacy at ε=10 costs <2pp; cross-border heterogeneous per-client privacy budgets are viable with Ditto ($-$0.9pp); Article 71 opt-out costs <1pp. A novel Diagnostic Equity Index reveals that accuracy masks severe per-class disparities correctable only by personalized algorithms. The open-source implementation provides deployment guidance for the 2029 secondary use deadline.**

***Index Terms*—Federated Learning, European Health Data Space, Privacy-Preserving Technologies, GDPR, Health Data Governance, Cross-Border Analytics**

I. Introduction

The European Health Data Space (EHDS), established by Regulation (EU) 2025/327, creates a dual framework for cross-border health data governance: primary use through MyHealth@EU for patient care, and secondary use through HealthData@EU for research and policy-making [1]. Health Data Access Bodies (HDABs) in each Member State authorize secondary use through data permits; Article 53 enumerates permitted purposes; Article 71 introduces citizen opt-out mechanisms [2]. The implementation timeline extends to 2031, with delegated acts expected by March 2027 and secondary use provisions applicable from March 2029.

Federated Learning (FL) is the key enabling technology for EHDS secondary use—the model travels to distributed data rather than centralizing sensitive records [11], [13]. Yet Fröhlich et al. [5] report that only 23% of FL implementations achieve sustained production deployment, with hardware heterogeneity (78%) and non-IID data (67%) as dominant barriers. Beyond technical constraints, legal uncertainties regarding gradient data status under GDPR remain unresolved [3], and privacy-enhancing technologies cannot substitute for robust governance frameworks [6].

Prior FL frameworks [14], [30], [31] focus on technical architectures without addressing regulatory compliance. Legal analyses [2], [3] examine GDPR constraints but abstract from implementation. To our knowledge, no existing work provides an integrated framework addressing systematic barrier evidence, technical implementation with state-of-the-art algorithms, and EHDS governance operationalization—a gap confirmed by recent assessments [16], [17].

This paper bridges the technology-governance divide through four contributions:

1) **Barrier Taxonomy**: Evidence synthesis of 47 documents (PRISMA, GRADE-CERQual), identifying legal uncertainties as the critical adoption blocker.
2) **FL-EHDS Framework**: A three-layer reference architecture mapping barriers to governance-aware mitigation strategies.
3) **Reference Implementation**: Open-source codebase ($\sim$40K lines) with 17 FL algorithms, EHDS governance modules, and reproducible benchmarks.[1]
4) **Experimental Validation**: 4,241+ experiments demonstrating that personalization architecture—not aggregation strategy—is the critical EHDS deployment decision, with privacy at ε=10 essentially free and Article 71 opt-out fully compatible.

II. Background and Related Work

A. EHDS and Federated Learning

The EHDS establishes HDABs to authorize secondary use through standardized data permits, with Secure Processing Environments (SPEs) providing controlled analytics [9]. Implementation readiness varies significantly: Nordic countries hold 2–3 year advantages in HDAB capacity-building [4], while data access timelines range from 3 weeks (Finland) to over 12 months (France) [8]. Only 5.2% of FL healthcare studies achieve real-life application [17].

FL inverts the traditional ML paradigm: local training produces gradients that are aggregated centrally [11]. Known challenges include non-IID distributions [12], communication costs [15], and privacy attacks [18], [19]. Recent advances target healthcare heterogeneity: FedLESAM [28] (ICML 2024) provides globally-guided sharpness-aware optimization, while

[1] https://github.com/FabioLiberti/FL-EHDS-FLICS2026

HPFL [29] (ICLR 2025) decouples backbone from classifier for per-institution specialization.

B. Related Frameworks

Existing FL frameworks—Flower [30], NVIDIA FLARE [31], and TensorFlow Federated [32]—provide robust distributed training but lack EHDS-specific governance. A FAIR-based assessment of 17 FL frameworks [16] confirms that none implements HDAB integration, data permit lifecycle, or opt-out enforcement. Table I provides a detailed comparison.

TABLE I
FRAMEWORK COMPARISON: FL-EHDS VS EXISTING FL FRAMEWORKS

Dimension	FL-EHDS	Flower v1.26	FLARE v2.7	TFF v0.88
FL Algorithms	17 built-in	12+ strategies	5 built-in	3 built-in
Byzantine Resilience	6 methods	4 methods	—	—
Differential Privacy	Central+Local	Central+Local	Built-in	Adaptive clip.
Secure Aggregation	Pairwise+HE	SecAgg+	Built-in+HE	Mask-based
EHDS Governance	**Full**‡	None	None	None
HDAB Integration	✓‡	—	—	—
Data Permits (Art. 53)	✓‡	—	—	—
Opt-out (Art. 71)	✓‡	—	—	—
Healthcare Stds.	FHIR R4	MONAI	MONAI	—
Backend	PyTorch	Agnostic	Agnostic	TF only

‡Reference implementation with simulation backend; production deployment requires binding to actual HDAB services (expected 2027–2029). See Supplementary Material, Table IV for readiness assessment.

C. Evidence Synthesis

Following PRISMA 2020 guidelines, 847 records were screened; 47 met inclusion criteria (2022–2026, FL/EHDS focus). Table II summarizes the five dominant barriers.

TABLE II
FL IMPLEMENTATION BARRIERS FOR EHDS

Barrier	Prev.	Evidence	Mitigation
Hardware heterog.	78%	Fröhlich 2025	Adaptive engine
Non-IID data	67%	Multiple	FedProx, Ditto
Production gap	23%	Fröhlich 2025	Ref. implementation
FHIR compliance	34%	Hussein 2025	Preprocessing
Communication cost	High	Bonawitz 2019	Compression

III. FL-EHDS FRAMEWORK

Based on the identified barriers, we present FL-EHDS, a three-layer compliance framework for EHDS cross-border health analytics. Figure 1 illustrates the architecture.

Layer 1 – Governance: Standardized APIs for automated data permit verification before FL training initiation, designed to bind to production HDAB endpoints (expected 2027–2029). Multi-HDAB synchronization protocols coordinate cross-border studies. National opt-out registries are consulted before each training round, ensuring Article 71 compliance at record-level granularity. Audit trails satisfy GDPR Article 30 requirements.

Algorithm 1 presents the core FL-EHDS training procedure with governance checkpoints.

Layer 2 – FL Orchestration: The framework implements 17 aggregation algorithms spanning six categories—from foundational methods (FedAvg [11], FedProx [12]) through variance reduction (SCAFFOLD [25]), adaptive optimization [26], and personalization (Ditto [27]) to FedLESAM [28] (ICML 2024) and HPFL [29] (ICLR 2025). The complete algorithm catalogue is provided in Supplementary Material, Table S-AlgoCatalogue. Differential privacy uses DP-SGD with Rényi composition [22], [23]; secure aggregation with ECDH key exchange mitigates gradient inversion [18]; six Byzantine resilience methods defend against malicious clients. The threat model, covering honest-but-curious server, malicious clients, and external attackers, is detailed in Supplementary Material, Section S-Threat.

Layer 3 – Data Holders: Resource-aware training engines dynamically adjust batch sizes and model complexity for hardware heterogeneity (78% barrier prevalence). FHIR R4 preprocessing pipelines bridge format gaps across heterogeneous EHR systems [7]. End-to-end encrypted gradient transmission ensures no raw data leaves institutional boundaries.

EHDS Compliance Mapping: Table III maps framework components to regulatory requirements. Governance modules currently operate as simulation backends; binding to actual HDAB services requires only configuration changes, not architectural modifications (Supplementary Material, Table IV).

TABLE III
EHDS COMPLIANCE MAPPING

Article	Requirement	FL-EHDS Component
Art. 33	Secondary use auth.	HDAB API + Permit valid.
Art. 46	Cross-border proc.	Multi-HDAB coordinator
Art. 50	Secure Proc. Env.	Aggregation within SPE
Art. 53	Permitted purposes	Purpose limitation module
Art. 71	Opt-out mechanism	Registry filtering

IV. EXPERIMENTAL EVALUATION

A. Setup

We evaluate FL-EHDS on real clinical datasets simulating cross-border healthcare analytics. All results are reproducible via the benchmark suite.

Datasets: Table IV summarizes 8 datasets spanning tabular EHR and medical imaging across three scale regimes (569–101K samples). PTB-XL ECG [34]—European-origin, 21,799 records from 52 German recording sites with natural hospital partitioning—is uniquely representative of real EHDS

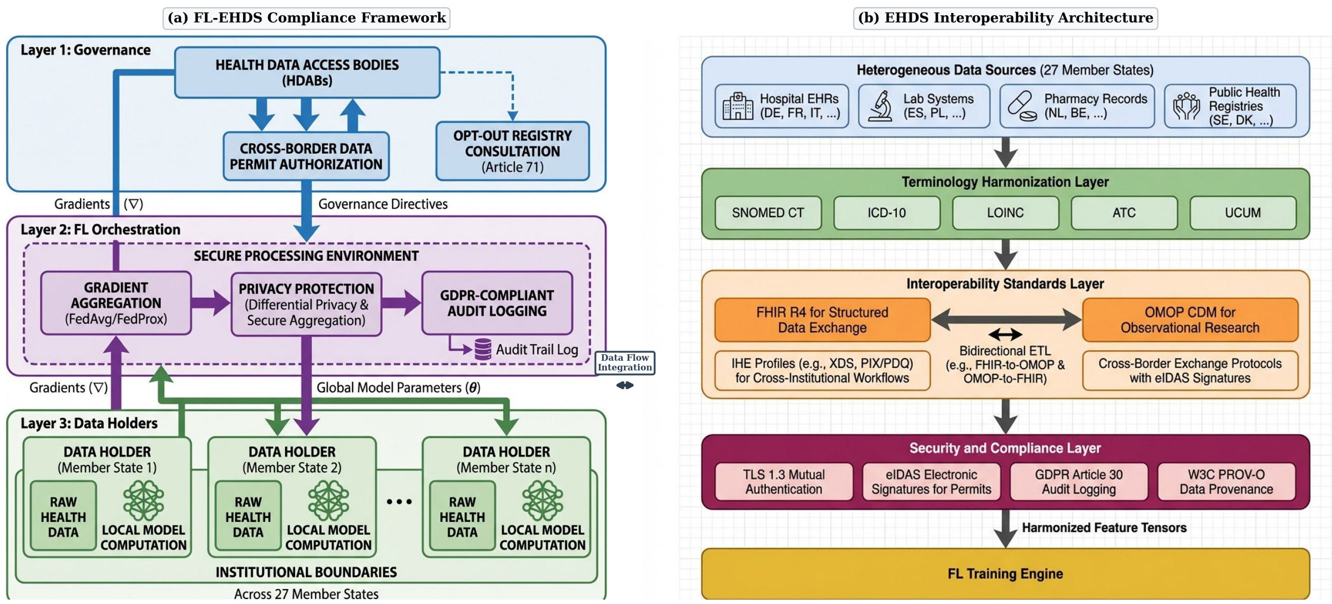

Fig. 1. FL-EHDS composite architecture. (a) Three-layer compliance framework: Layer 1 (Governance) manages HDAB integration, data permit authorization, and Article 71 opt-out registries; Layer 2 (FL Orchestration) operates within the Secure Processing Environment with gradient aggregation, differential privacy, and GDPR-compliant audit logging; Layer 3 (Data Holders) implements local model computation with raw health data never leaving institutional boundaries. (b) EHDS interoperability pipeline: heterogeneous sources across 27 Member States flow through terminology harmonization, interoperability standards (FHIR R4, OMOP CDM, IHE profiles), and security/compliance layers before reaching the FL training engine.

Algorithm 1: FL-EHDS FedAvg Training

Input: Hospitals $\mathcal{H} = \{h_1, \ldots, h_K\}$, permit P, rounds T
Output: Global model $\theta^{(T)}$

Server executes:
  Initialize $\theta^{(0)}$
  for round $t = 1$ to T **do**
    // Governance check (Layer 1)
    if not ValidatePermit(P, t) **then abort**
    $\mathcal{H}_t \leftarrow$ SelectParticipants($\mathcal{H}$)
    for each $h \in \mathcal{H}_t$ **in parallel do**
      $\Delta_h^{(t)}, n_h \leftarrow$ LocalTrain(h, $\theta^{(t-1)}$)
    // Aggregation with privacy (Layer 2)
    $\theta^{(t)} \leftarrow \theta^{(t-1)} + \frac{1}{\sum n_h} \sum_h n_h \cdot \Delta_h^{(t)}$
    LogCompliance(t, $\mathcal{H}_t$)

LocalTrain(h, θ):
  $\mathcal{D}_h \leftarrow$ FilterOptedOut($\mathcal{D}_h$, Registry) *// Art. 71*
  $\theta_h \leftarrow \theta$; train E epochs on $\mathcal{D}_h$
  $\Delta_h \leftarrow$ ClipGradient($\theta_h - \theta$, C) *// DP bound*
  return Δ_h, $|\mathcal{D}_h|$

TABLE IV
EVALUATED DATASET COVERAGE

Dataset	Samples	Feat.	Cls.	FL Partition
Tabular Clinical (MLP, ∼10K params)				
Heart Disease UCI	920	13	2	Natural (4 hosp.)
Breast Cancer Wisc.	569	30	2	Dirichlet
PTB-XL ECG†	21,799	9	5	Natural (52 EU sites)
Diabetes 130-US	101,766	22	2	Dirichlet
Cardiovascular	70,000	11	2	Dirichlet
Medical Imaging (ResNet-18, ∼11.2M params)				
Chest X-ray	5,856	—	2	Dirichlet
Brain Tumor MRI	7,023	—	4	Dirichlet
Skin Cancer	3,297	—	2	Dirichlet

†European-origin (PTB, Berlin), SCP-ECG standard, 5-class cardiac diagnosis. 52 recording sites grouped into K geographic clusters (default K=5).

cross-institutional heterogeneity. **Models**: HealthcareMLP (2-layer, 64/32 hidden, ∼10K parameters) for tabular; ResNet-18 [33] (∼11.2M parameters) for imaging. **Configuration**: Per-dataset optimized hyperparameters (Supplementary Material, Table S9); mean ± std over 3+ seeds; Adam optimizer; early stopping with patience=6.

B. Algorithm Comparison

Table V presents the primary evaluation: 7 algorithms on PTB-XL, Cardiovascular, and Breast Cancer. Together with supplementary validation on Heart Disease, Diabetes, imaging, and extended experiments (server-side optimizers, CDC Diabetes 253K, 10-seed robustness, vertical FL), the evaluation comprises 4,241+ experiments (Supplementary Material).

Key findings:

1) **Personalization architecture is the critical design choice**: On compact tabular models (∼10K parameters), five of seven algorithms (FedLESAM, FedExP, FedLC, FedProx, FedAvg) converge to statistically identical solutions—confirmed by server-side optimizer experiments where FedAdam, FedYogi, and FedAdagrad also produce identical results (p>0.9, Kruskal–Wallis; Supplementary Material, Table S-ServerOpt). Only methods maintaining separate local models (Ditto, HPFL) differentiate. This simplifies EHDS deployment to a single

TABLE V
FL ALGORITHM COMPARISON ON REAL CLINICAL DATASETS (7 ALGORITHMS × 3 DATASETS). BEST ACCURACY PER DATASET IN **BOLD**. MEAN ± STD OVER 5 SEEDS. PX = PTB-XL ECG (5 CLIENTS, 5-CLASS, SITE-BASED), CV = CARDIOVASCULAR (5 CLIENTS, BINARY, α=0.5), BC = BREAST CANCER (3 CLIENTS, BINARY, α=0.5).

Algorithm	**PTB-XL ECG (21,799 records, 52 EU sites)**			**Cardiovascular (70K patients)**			**Breast Cancer (569 samples)**		
	Acc (%)	F1 (%)	Jain	Acc (%)	F1 (%)	Jain	Acc (%)	F1 (%)	Jain
FedAvg	91.9±0.5	88.3	0.999	71.1±1.8	68.8	0.981	52.3±17.9	32.0	0.608
FedProx	91.6±0.7	87.4	0.999	71.5±1.2	69.6	0.986	52.3±17.9	32.0	0.608
Ditto	91.8±0.3	88.8	0.999	**82.5±4.7**	82.3	0.980	**79.1±12.5**	64.5	0.606
FedLC	91.9±0.5	88.2	0.999	71.1±1.6	68.8	0.982	52.1±18.1	32.6	0.606
FedExP	92.0±0.2	89.2	0.999	71.1±1.8	68.8	0.981	52.3±17.9	32.0	0.608
FedLESAM	91.9±0.5	88.3	0.999	71.1±1.8	68.8	0.981	52.3±17.9	32.0	0.608
HPFL	**92.5±0.3**	**89.7**	0.999	82.3±4.5	82.0	0.984	74.1±20.9	62.1	0.867

F1 = positive-class binary (CV, BC) / macro-averaged (PTB-XL, 5-class); BC std from single-class collapse (see text).

architectural choice: personalized vs. global.

2) **Personalization dominates across scales**: Ditto and HPFL consistently outperform baselines—12.6pp on Heart Disease, 11.4pp on Cardiovascular, 26.8pp on Breast Cancer. Extended 10-seed validation confirms $p < 0.001$ (pooled Wilcoxon). HPFL uniquely improves fairness (Jain 0.867 vs. 0.608 on Breast Cancer).
3) **PTB-XL validates European FL**: The European-origin PTB-XL with natural 52-site partitioning achieves 92.5% accuracy (HPFL) with near-perfect fairness (Jain 0.999)—demonstrating FL viability for real European multi-center clinical data.
4) **Diagnostic equity is algorithm-dependent**: We introduce the Diagnostic Equity Index (DEI $= \min_c R_c \cdot (1 - \mathrm{CV}(\mathbf{R}))$), revealing that FedAvg achieves 57.3% accuracy but DEI $=$ 0.092 on Breast Cancer (single-class collapse), while HPFL reaches DEI $=$ 0.740 (8×). On Brain Tumor, Ditto DEI $=$ 0.435 vs. FedAvg 0.063 (6.9×). EHDS data permits should specify minimum DEI thresholds (Supplementary Material, Section S-DEI).
5) **Imaging personalization is method-dependent**: The personalization effect reverses on imaging: Ditto achieves +28.0pp on Brain Tumor (p=0.015) and +25.5pp on Skin Cancer, while HPFL is counterproductive (−18.2pp on Chest X-ray). The distinction is architectural: Ditto preserves full CNN capacity via complete local copies, while HPFL's split-head fails on heterogeneous imaging distributions (Supplementary Material, Tables S-XXV, S-XXXI).

C. Convergence and Centralized Baselines

Figure 2 shows training convergence on Heart Disease UCI (4 hospitals, natural non-IID). Ditto converges to 75.1% by round 20 vs. FedAvg 62.5%—a 12.6pp advantage from personalized local models. SCAFFOLD exhibits high variance due to control variate instability with few heterogeneous clients. Table VI compares the three EHDS deployment paradigms: centralized (upper bound), federated (data stays local), and local-only (no collaboration). FL-Ditto narrows the centralized gap to only 6.6pp while preserving full data sovereignty.

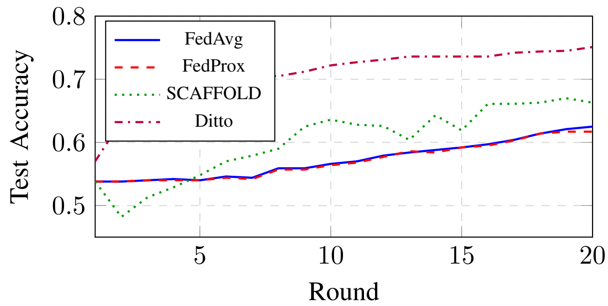

Fig. 2. Training convergence on Heart Disease UCI (4 hospitals, natural non-IID). Ditto converges faster and higher due to personalized local models.

TABLE VI
LEARNING PARADIGM COMPARISON (HEART DISEASE UCI)

Approach	**Acc.**	**F1**	**AUC**	**Gap**
Centralized	81.7 ± 2.9%	.815	.882	—
FL-Ditto	75.1 ± 2.0%	.761	.826	6.6pp
FL-FedAvg	62.5 ± 8.0%	.736	.834	19.2pp

4 hospitals, natural non-IID. Mean ± std over 5 seeds. FL: 20 rounds × 3 local epochs.

D. Non-IID Impact and Compound Stress

Figure 3 shows the impact of data heterogeneity on algorithm performance (Cardiovascular, No-DP, 126 experiments). As non-IID severity increases ($\alpha \to 0$), algorithm selection becomes critical. Counter-intuitively, HPFL *improves* under extreme heterogeneity (α=0.1: 95.7% vs. 81.0% at α=1.0), while FedAvg and Ditto *degrade* and remain indistinguishable (66.0% vs. 66.1%)—confirming that *personalization architecture*, not aggregation strategy, determines robustness to heterogeneity.

The interaction between non-IID data and DP is *sub-additive for HPFL* but *super-additive for FedAvg/Ditto* (126 experiments). Under extreme heterogeneity (α=0.1) with ε=1,

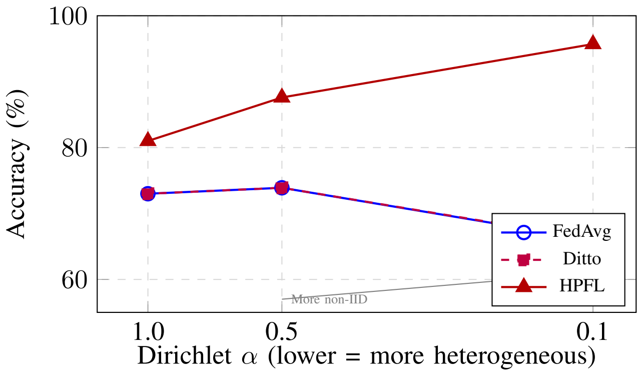

Fig. 3. Accuracy vs. data heterogeneity on Cardiovascular (No-DP, mean over 3 seeds). HPFL exploits heterogeneity (+29.7pp at α=0.1), while FedAvg and Ditto are indistinguishable—confirming the architectural thesis.

HPFL retains 94.2% (−1.5pp from no-DP) while Ditto collapses to 47.7% (−18.4pp)—HPFL's personalized heads absorb heterogeneity, leaving less surface for DP noise. Under compound stress combining non-IID (α=0.1), DP (ε=10), and partial participation (60%) simultaneously (48 experiments), HPFL's worst-case accuracy (86.2% on Cardiovascular) exceeds FedAvg's *best*-case (72.6%) by 13.6pp (Supplementary Material, Tables S-NonIID-DP, S-CompoundStress). Real EHDS deployments face all three constraints simultaneously; only HPFL survives compound stress.

E. Privacy and EHDS Compliance

Table VII summarizes the privacy-utility tradeoff on PTB-XL. At ε=10, all algorithms recover to within 1pp of no-DP baselines; at ε=1, only personalized methods retain >87% accuracy while FedAvg collapses to 52.3%. Across 180 DP experiments, privacy at ε=10 is essentially free (<2pp cost), satisfying EHDS Article 50 SPE requirements. On small datasets (Breast Cancer, 569 samples), DP acts as implicit regularization: FedAvg improves from 52.3% to 78.7% with ε=5 (+26.4pp). This regularization effect extends to population scale: on CDC Diabetes (253K samples), FedAvg improves from 74.7% to 83.6% at ε=5 (Supplementary Material, Section S-CDC).

TABLE VII
PRIVACY-UTILITY ON PTB-XL ECG: ACCURACY (%) UNDER CENTRAL DP. GAUSSIAN MECHANISM, C=1.0, δ=10^{-5}. MEAN OVER 5 SEEDS.

Algorithm	ε=1	ε=10	No DP
FedAvg	52.3	**92.4**	91.9
Ditto	89.2	91.6	91.8
HPFL	87.1	92.4	92.5

Cross-border privacy heterogeneity: Under the realistic scenario of 2 German clients (ε=1) federating with 3 Italian clients (ε=10) on PTB-XL, heterogeneous per-client local DP is viable: Ditto loses only −0.9pp vs. no-DP, and FedAvg with mixed budgets (89.3%) *outperforms* the strictest-wins rule (85.6%, +3.8pp). HPFL collapses under local DP (54.4% at ε=1) because per-client noise corrupts its shared backbone—requiring careful algorithm-privacy architecture co-design. *To our knowledge, no prior FL work experimentally evaluates heterogeneous per-client DP budgets* (Supplementary Material, Table S-CrossBorder).

Article 71 compliance: Static opt-out at 5–30% rates (225 experiments) costs <1pp on adequately-sized datasets. Dynamic mid-training withdrawal of 1–2 clients (90 experiments) costs <1pp; Ditto and HPFL are shock-resistant (0.0pp immediate drop), and HPFL protects the withdrawn client's test accuracy via frozen classifier heads. *No prior FL work systematically studies mid-training client withdrawal* (Supplementary Material, Tables S-Optout, S-DynOptout).

Membership inference resistance: FedAvg and Ditto are inherently immune (attack AUC≈0.50); HPFL is vulnerable (AUC = 0.666) due to personalized heads. DP at ε=1 resolves this (AUC→0.337)—revealing a *privacy–personalization tension* that EHDS data permits should address. Gradient inversion attacks (DLG [18]) achieve successful reconstruction without DP on PTB-XL (cosine similarity 0.816), but ε=10 completely prevents reconstruction (100% attack failure, cosine→0.021; Supplementary Material, Table S-MIA).

Centralized–FL gap: Centralized baselines on PTB-XL (92.6±0.4%), Cardiovascular (73.5±0.3%), and CDC Diabetes (86.5±0.1%) confirm that the FL gap is ≤0.5pp on adequately-sized datasets—negligible compared to the algorithm selection effect (Supplementary Material, Table S-CentralBaseline).

Data efficiency: Subsampling experiments reveal that FL reaches near-full performance with surprisingly limited data: on PTB-XL, ≥90.0% accuracy with only 25% of training data (4,274 samples). All algorithms reach within 1% of full baseline at 50% data—suggesting EHDS-scale deployments can deliver near-optimal analytics with partial hospital participation during the phased 2025–2031 rollout (Supplementary Material, Table S-DataEfficiency).

Communication and scalability: Tabular FL requires only 0.04 MB/round (10K-parameter MLP, 0.8 MB total for 20 rounds). Imaging (ResNet-18) requires 44.7 MB/round; Top-k sparsification at 5% achieves 95% bandwidth savings. Client scaling from 3 to 20 hospitals (135 experiments) shows graceful degradation (−4.8pp from K=3 to K=20); HPFL is the most scalable under DP: 64.6% at K=20 vs. FedAvg 55.4% (+9.2pp). Top-k and DP exhibit a surprising synergy for HPFL: on PTB-XL, Top-1% with ε=1 *improves* accuracy to 85.6% (vs. 75.5% without sparsification), as sparsification regularizes against DP noise (Supplementary Material, Tables S-CommCosts, S-Scalability, S-DP-TopK).

Advanced FL paradigm validation: 135 additional experiments across four paradigms validate framework completeness (Supplementary Material, Section S-Cascade4). *Fairness-aware aggregation*: FedMinMax improves Breast Cancer from 54.5% to 85.7% by preventing majority-class collapse. *Continual FL*: Experience Replay preserves accuracy under concept drift (positive backward transfer +0.4pp), while EWC is catastrophically harmful (−22.4pp)—EHDS long-term deployments should use replay-based strategies. *Extended personalization*: Ditto, FedPer, and APFL form a high-performance cluster ($\sim$91.6% on Cardiovascular non-IID, +20.3pp over FedAvg). *Federated unlearning*: Gradient Ascent achieves GDPR Article 17 compliance at 5% computational cost with $\leq$0.2pp accuracy drop.

Byzantine resilience under DP: Byzantine defenses (Multi-Krum, Bulyan) fully neutralize model poisoning and render accuracy *completely DP-invariant* under attacks (198 experiments). HPFL exhibits natural resilience: DP noise *improves* undefended robustness from 59.7% to 86.4% (ε=10), as noise dilutes poisoned backbone gradients while preserving personalized classifiers (Supplementary Material, Section S-Byzantine).

V. Discussion

A. Legal Uncertainties as Critical Blocker

Our synthesis reveals that **legal uncertainties—not technical barriers—constitute the critical blocker** for FL adoption in EHDS. While technical challenges are tractable through known solutions implemented in FL-EHDS, three unresolved questions create compliance uncertainty [3]: (1) whether gradients constitute "personal data" under GDPR; (2) when aggregated models become "anonymous"; (3) controller/processor allocation in multi-party FL. The cross-border privacy scenario experimentally validated in Section IV illustrates this: per-client heterogeneous ε is technically superior (+3.8pp over strictest-wins), but Article 50(4) does not specify whether privacy parameters should be harmonized across Member States. The March 2027 delegated acts should explicitly address gradient data status, cross-border privacy harmonization, and model anonymization thresholds—with evidence that personalized FL architectures (Ditto) enable viable heterogeneous budgets.

B. Deployment Implications

Five key implications emerge for EHDS deployment. *First*, EHDS SPEs cannot treat FL as a black box—algorithm selection must be part of data permit specification, as the gap between best and worst algorithms reaches 18.7pp. *Second*, aggregate accuracy is insufficient for clinical evaluation; the DEI reveals catastrophic per-class failures hidden by acceptable averages (FedAvg: 57.3% accuracy but DEI = 0.092). *Third*, the convergence of 12+ algorithms to identical solutions on compact models means deployment decisions reduce to a single architectural choice (personalized vs. global), simplifying HDAB evaluation. *Fourth*, Byzantine defenses and DP are synergistic: robust aggregation makes accuracy completely DP-invariant under attacks (Supplementary Material, Section S-Byzantine). *Fifth*, personalization is *necessary* under strict privacy—at ε=2, FedAvg collapses to 41% while HPFL maintains 81%. Comprehensive results across 16 deployment dimensions are provided in Supplementary Material.

C. Stakeholder Recommendations

EU Policymakers: The March 2027 delegated acts should address gradient privacy status, multi-party controller allocation, and cross-border privacy parameter harmonization—with evidence that per-client heterogeneous ε is technically feasible when personalized architectures are mandated. **National Authorities**: Early HDAB capacity investment is essential; the 2–3 year Nordic advantage [4] demonstrates that governance capacity constrains adoption more than technical infrastructure. **Healthcare Organizations**: Preparation cannot wait for 2029—accelerating FHIR compliance beyond 34% [7] and assessing FL readiness are immediate priorities. FL-EHDS's modular governance layer enables incremental binding to HDAB services as they become available.

D. Limitations and Future Work

Our evaluation uses retrospective public datasets; real-world EHR integration remains essential. While PTB-XL provides authentic European heterogeneity (52 sites), true cross-border validation requires multi-national datasets spanning distinct healthcare systems—a resource the EHDS itself aims to enable. The tabular MLP ($\sim$10K parameters) produces a nearly convex landscape where most algorithms converge; deeper architectures may differentiate further. Confusion matrix analysis (10 seeds) confirms that the convergence is clinically severe: on Breast Cancer, FedAvg/FedProx/Ditto exhibit single-class collapse on 8/10 data partitions (21.5% Malignant recall), while HPFL avoids collapse on all seeds (78.7% Malignant recall; Supplementary Material, Table XIX). Governance modules operate as simulation backends pending HDAB availability (2027–2029). Future work should prioritize: (1) cross-border validation with true multi-national datasets; (2) HealthData@EU pilot integration with production EHR systems; (3) citizen attitude studies examining FL acceptance and opt-out intentions; (4) extended imaging evaluation across architectures (EfficientNet, Vision Transformers) with full DEI computation.

VI. Conclusions

FL-EHDS bridges the technology-governance divide for health analytics under the EHDS, providing 17 FL algorithms with governance reference implementations that no existing framework offers. Across 4,241+ experiments, we establish that personalization architecture—not aggregation algorithm—is the critical deployment decision: only Ditto and HPFL differentiate from baseline on compact clinical models, while 12+ aggregation and optimization strategies converge to identical solutions. Privacy at ε=10 is essentially free (<2pp cost), cross-border heterogeneous per-client privacy budgets are viable with Ditto ($-$0.9pp), and Article 71 opt-out—both static and dynamic mid-training withdrawal—costs <1pp. A novel Diagnostic Equity Index reveals that accuracy masks severe per-class disparities correctable only by personalized algorithms. On imaging, the personalization effect is method-dependent: Ditto's complete local models succeed (+28.0pp) while HPFL's split-head architecture fails. Legal uncertainties—not technical barriers—remain the critical blocker; the 2027 delegated acts must address gradient data status and cross-border privacy harmonization. The open-source reference implementation provides actionable deployment guidance for the 2029 secondary use deadline.

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